



Retro Data

Azure SQL Databases, tales from the trenches

183.102

154.178

245.57



Database

Data Warehouse

Trenches

Setup





The old way

Choosing your own hardware
Standard or Enterprise Edition
Configuration

SQL Server



Azure SQL

The new way

Cloud service, Managed Instance or on a VM

What Tier to choose

What SKU to choose

Network connectivity challenges

No more SA!

Migration is... different

Where to look

Where to start
What to look out for



Opinions



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On premises

SQL Server

Choosing your own hardware
Standard or Enterprise Edition
Configuration



Azure SQL

Let's cloud!

Cloud service, Managed Instance or on a VM

What Tier to choose

What SKU to choose

Network connectivity challenges

No more SA!

Migration is... different

- ▶ Lift and shift or migrate?
- ▶ What Tier should I choose?
- ▶ Within the Tier, what SKU?
- ▶ Can I change between Tiers?
- ▶ Do I need to think about other things?

- ▶ Datawarehousing
- ▶ From on-premises to Azure
- ▶ New database



Key Performance Indicators

- ▶ Basic
- ▶ Standard
- ▶ Premium
- ▶ General Purpose (serverless or provisioned)
- ▶ Hyperscale
- ▶ Business Critical

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- ▶ General Purpose (serverless (< 183 hours per month) or provisioned)
- ▶ Hyperscale
- ▶ Business Critical

- ▶ Basic
- ▶ Standard
- ▶ Premium
- ▶ General Purpose (serverless or provisioned)
- ▶ Hyperscale Gen 5, Premium or Premium Memory Optimized and DC (and serverless)
- ▶ Business Critical

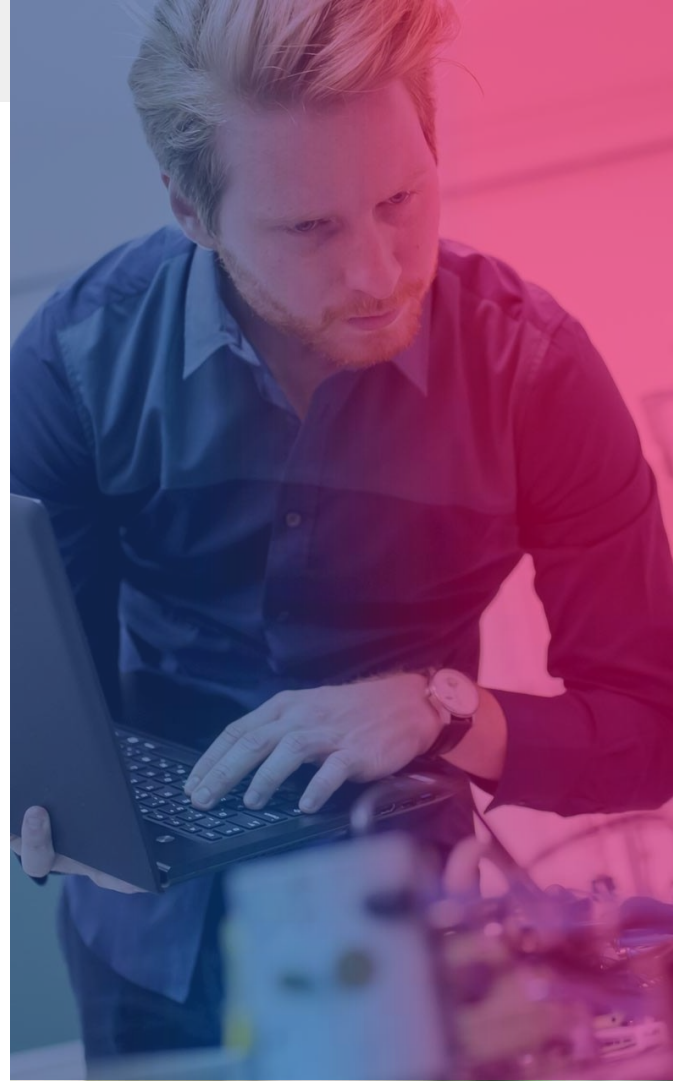
- ▶ Basic
- ▶ Standard
- ▶ Premium
- ▶ General Purpose (serverless or provisioned)
- ▶ Hyperscale
- ▶ Business Critical Gen 5, and DC

- ▶ Datawarehousing
- ▶ From AWS to Azure
- ▶ Migration from three VM's to one Managed Instance

Managed instance

- ▶ Hybrid Saas and PaaS
- ▶ General Purpose and Premium
- ▶ Almost like SQL on a machine, but not quite

How many are there?



Tiers and SKU's

Main configurations, excluding redundancy, IOPS, replica's and minimum cores

- | | |
|---|-------------|
| ▶ Hyperscale | 69 options |
| ▶ Managed Instance | 34 options |
| ▶ General Purpose | 38 options |
| ▶ Business Critical | 27 options |
| ▶ DTU (Basic, Standard, Premium) | 16 options |
| ▶ Total | 184 options |
| ▶ (and the one that shall not be named) | |

A top-down view of a diverse assortment of fresh produce, including a halved papaya with black seeds, a blood orange, a yellow tomato, a red bell pepper, a cantaloupe, artichokes, onions, and various small tomatoes, all arranged on a light blue surface. The text "SKU Differences" is prominently displayed in the center in a large, bold, yellow font.

SKU Differences

- ▶ DTU's (Database Transaction Units)
- ▶ Cores
- ▶ Minimum number of cores
- ▶ Database sizing
- ▶ TempDB files
- ▶ Storage
- ▶ Iops (Input Output Operations)
- ▶ Log Rate
- ▶ Replication and redundancy
- ▶ Backups

- ▶ Blend of CPU, Memory, reads and writes
- ▶ Calculator: <https://dtucalc.azurewebsites.net/>
- ▶ Total % Processor Time
 - Total disk reads per second
 - Total disk writes per second
 - Total log bytes flushed per second

- ▶ DTU's
- ▶ Cores
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Choices

So, we choose a Tier, choose the SKU and success?

- ▶ Yeah, well... no
- ▶ Start small
- ▶ Keep your KPI in check
- ▶ Keep your budget in check
- ▶ Don't be afraid to experiment
- ▶ Re-evaluate when load or volume changes

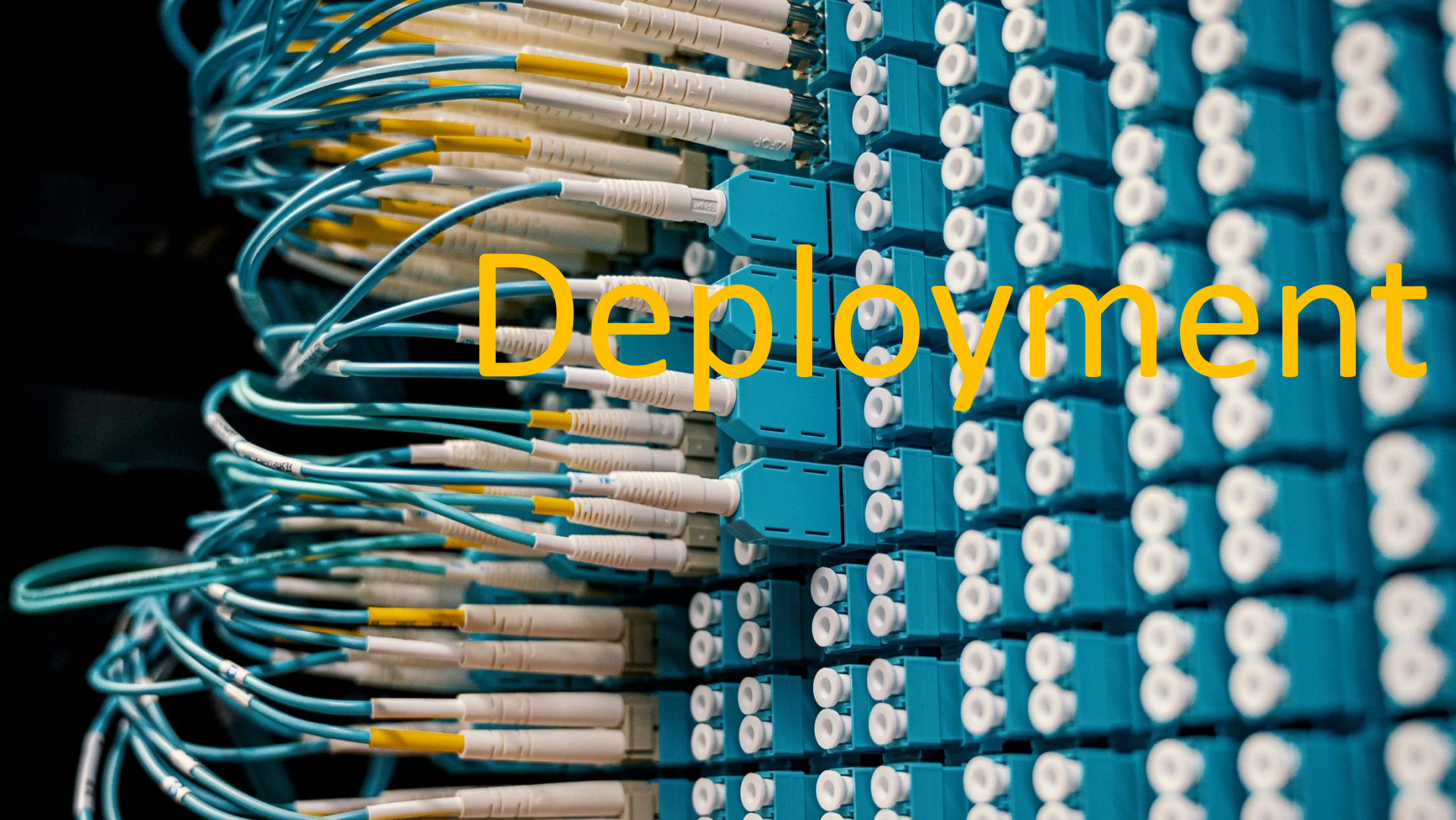
- ▶ On-premises twice as fast
- ▶ Analyse disk performance, wait stats and use Query Store
- ▶ Change tier and SKU
- ▶ Change ETL process
- ▶ Azure 10% quicker than on-premises
- ▶ Writes slow you down!



VM if you want to transfer
your workload to the cloud
without any changes

MI if you want to transfer your
workload to the cloud without
major changes but want to
take some advantages of
cloud offerings. Now also in
general and premium

PaaS if you want all the cloud
goodies with the least
management overhead



Deployment

- ▶ Connectivity
- ▶ Public or Private Endpoint
- ▶ Duration
- ▶ Network and Firewall

- ▶ Connectivity

- ▶ Public or Private Endpoint

- ▶ Duration

- ▶ Network and Firewall

- ▶ Connectivity
- ▶ Public or Private Endpoint
- ▶ Duration
- ▶ Network and Firewall

- ▶ Connectivity
- ▶ Public or Private Endpoint
- ▶ Duration (scaling takes time!)
- ▶ Network and Firewall

- ▶ Connectivity
- ▶ Public or Private Endpoint
- ▶ Duration
- ▶ Network and Firewall

Deployment

Connection Policy

Configure how clients communicate with your SQL database server.[Learn more](#)

Connection policy

- ☐ Default - Uses Redirect policy for all client connections originating inside of Azure (except for all client connections originating outside Azure)
- ☐ Proxy - All connections are proxied via the Azure SQL Database gateways
- ☒ Redirect - Clients establish connections directly to the node hosting the database

Encryption in transit

This server supports encrypted connections using Transport Layer Connections (TLS). Any login attempts from clients using a TLS version rejected. For information on TLS version and certificates, refer to connecting with TLS/SSL.[Learn more](#)

Minimum TLS version

TLS 1.2

- ▶ Connectivity
- ▶ Public or Private Endpoint
- ▶ Duration
- ▶ Network and Firewall

A close-up photograph of a green-painted wooden door. A heavy, rusty metal chain is draped across the door, secured by a brass padlock in the center. The door shows signs of wear, with some paint chipping and visible wood grain. A metal keyhole is visible in the lower right corner.

Security

- ▶ Azure AD Administrators group
 - ▶ Azure AD User group(s)
 - ▶ SQL Accounts
 - ▶ Outbound Firewall settings (data exfiltration)
 - ▶ Microsoft Defender and auditing
- It's YOU!!**

Where to look

Where to start
What to look out for



- ▶ Read the documentation
- ▶ Create a script
- ▶ Run it for baseline
- ▶ Run it against the Azure offerings
- ▶ Read the documentation

- ▶ <https://docs.microsoft.com/en-us/azure/azure-sql/database/resource-limits-vcore-single-databases?view=azuresql>
- ▶ <https://docs.microsoft.com/en-us/azure/azure-sql/database/resource-limits-dtu-single-databases?view=azuresql>

Setup



- ▶ Hardware check
- ▶ Insert
- ▶ Select
- ▶ Delete
- ▶ Update
- ▶ SqlStress to (over)load the database
- ▶ Databases as-is, no configuration or tuning

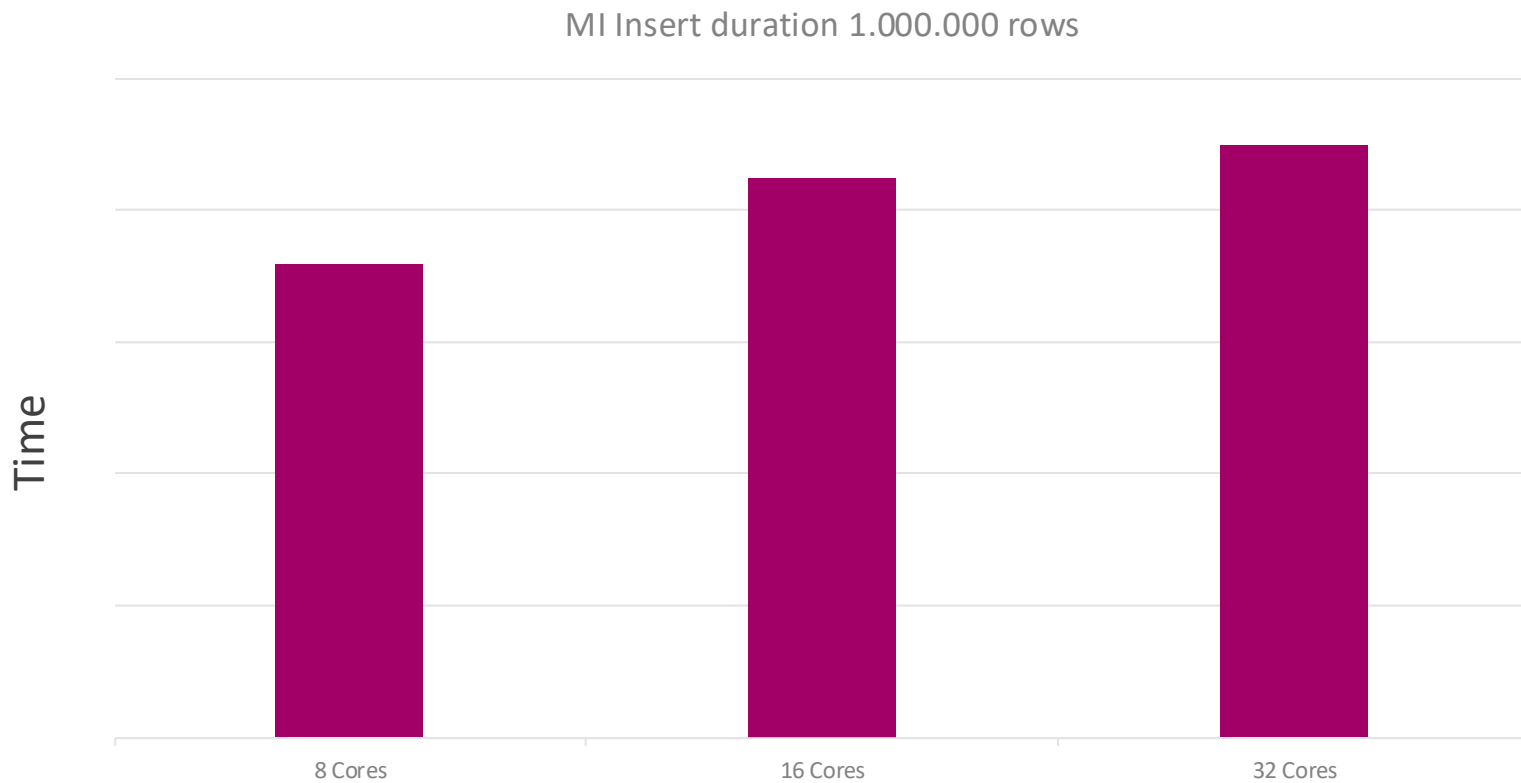




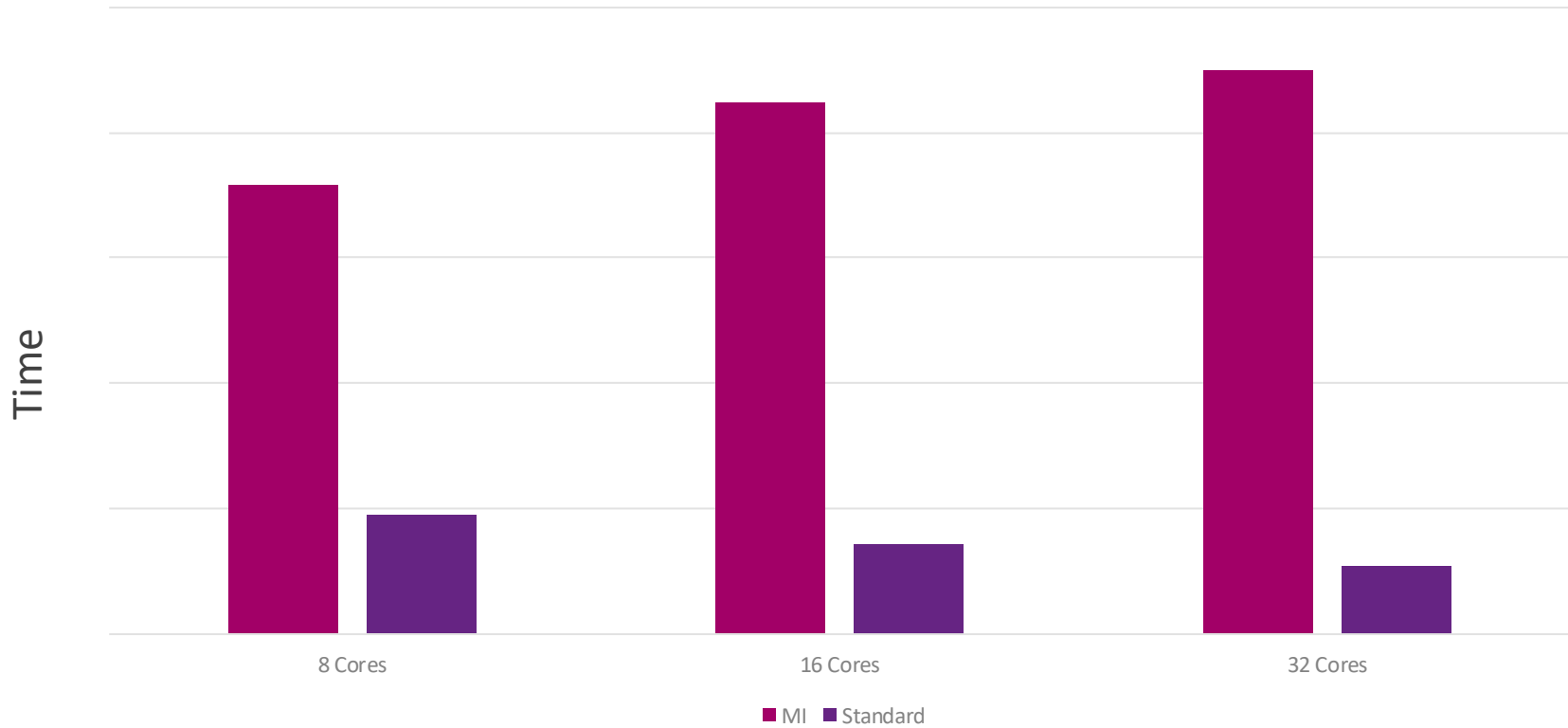
8, 16 and 32 core
environments



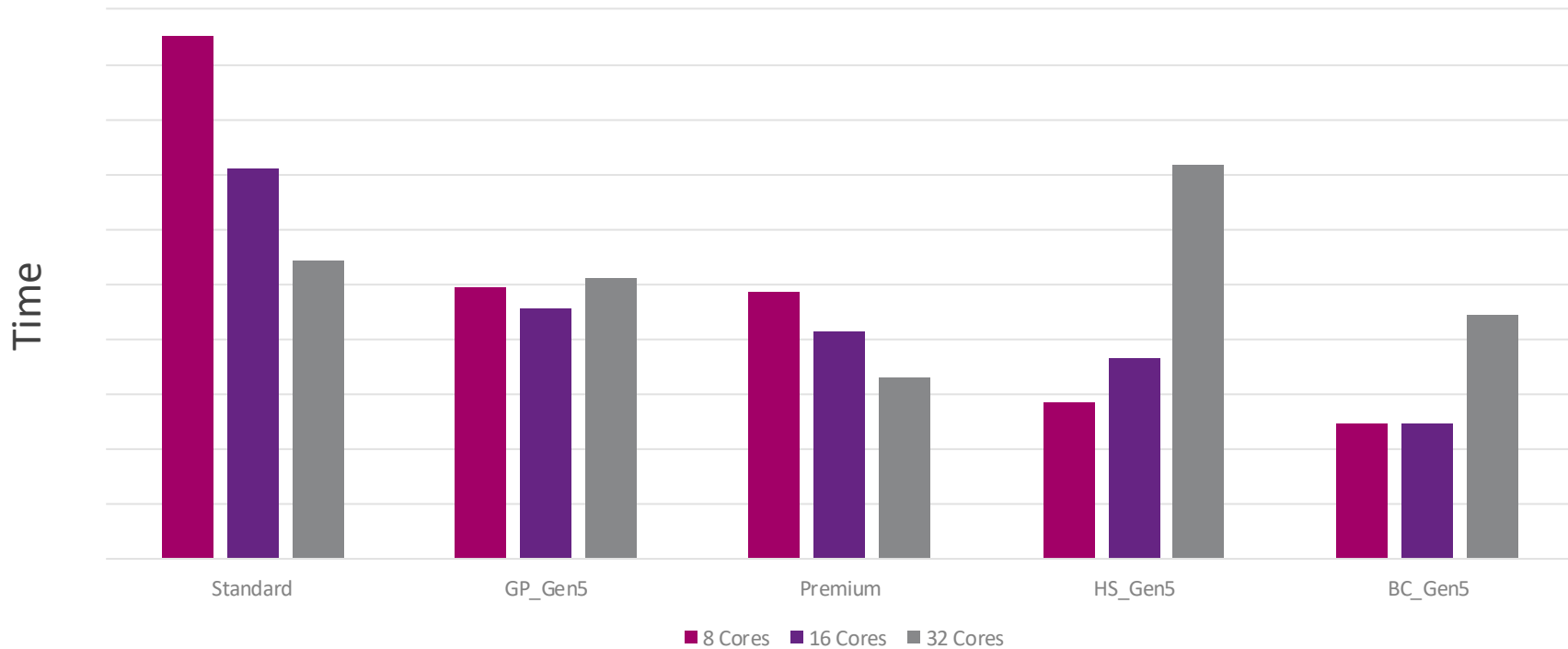
Per query graphs



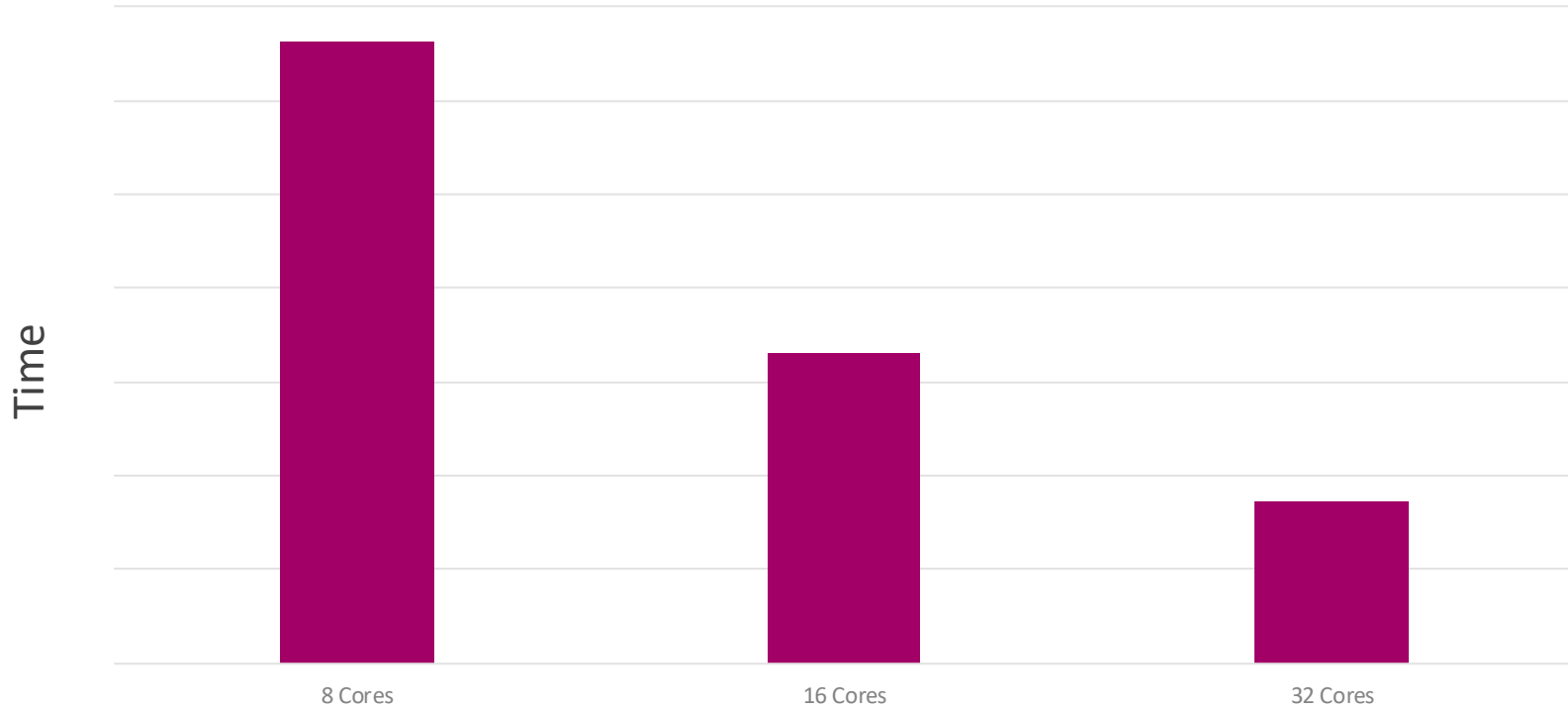
MI and Standard Insert duration 1.000.000 rows



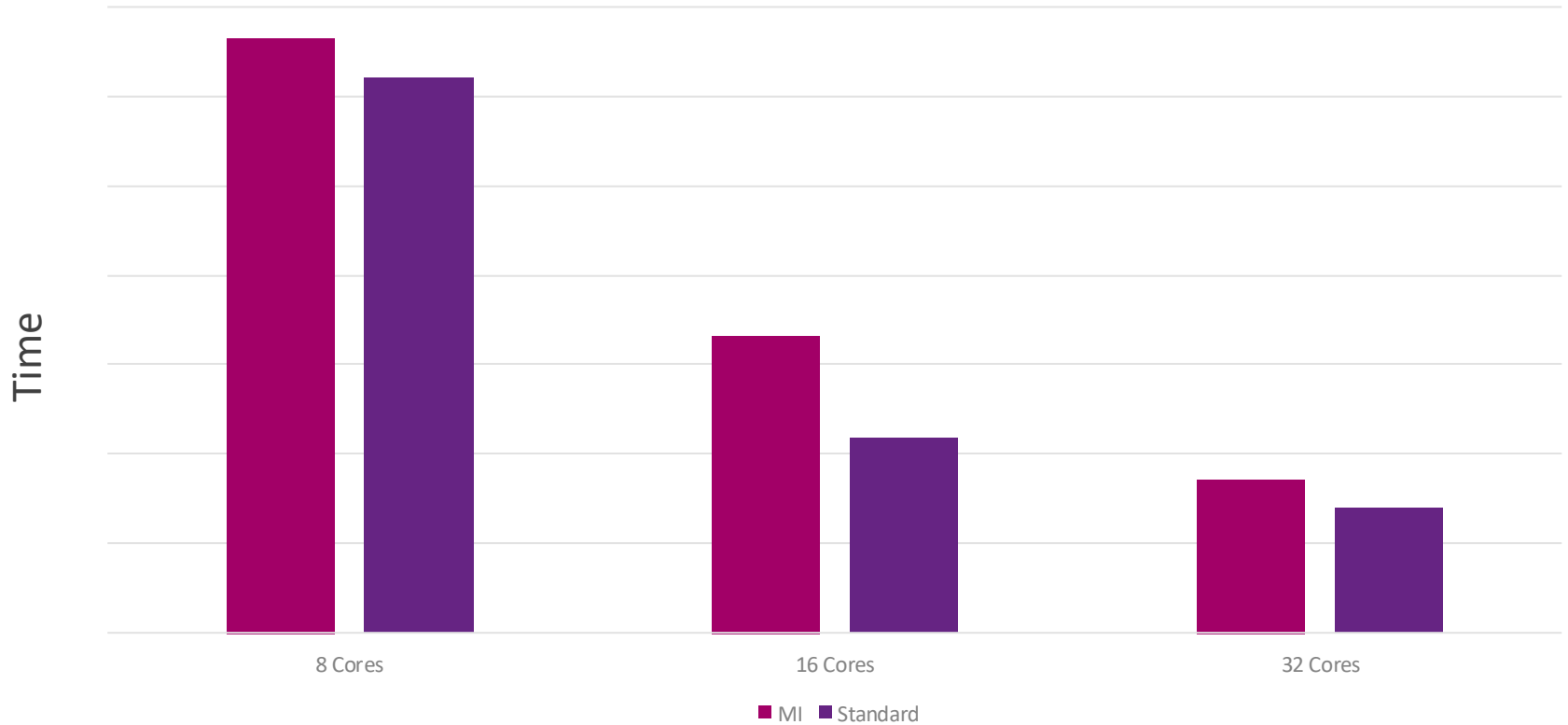
Duration insert 1.000.000 rows



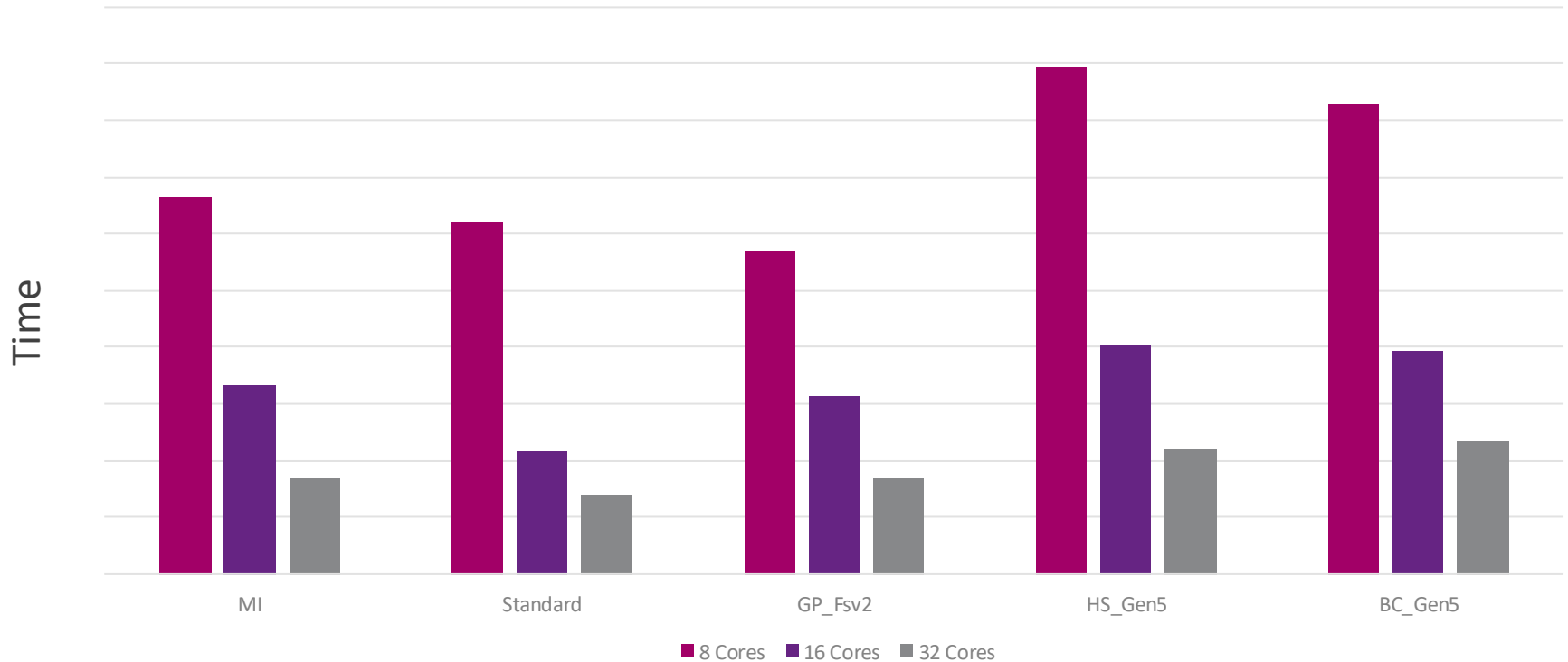
MI Duration Select 30.000 rows



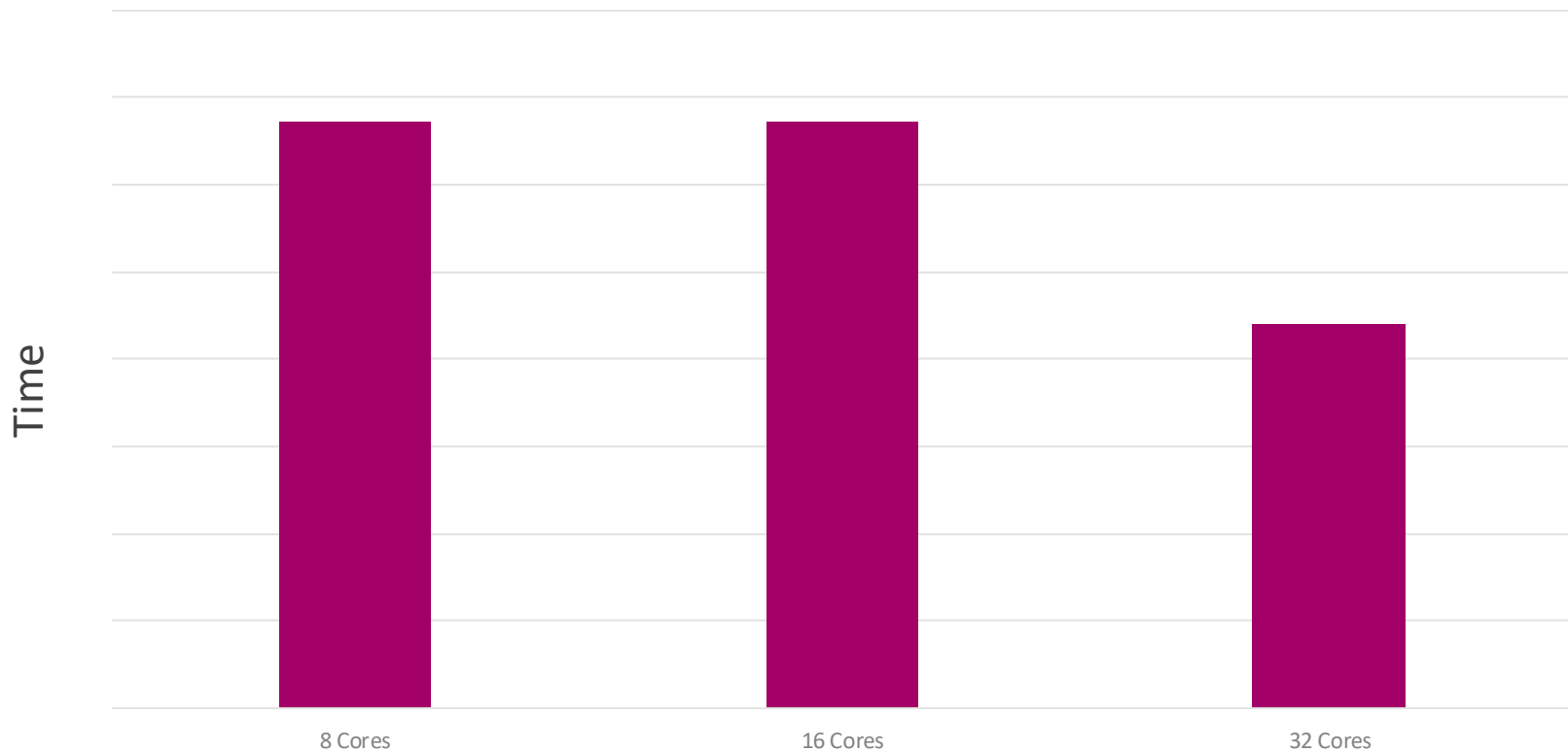
MI and Standard Duration Select 30.000 rows



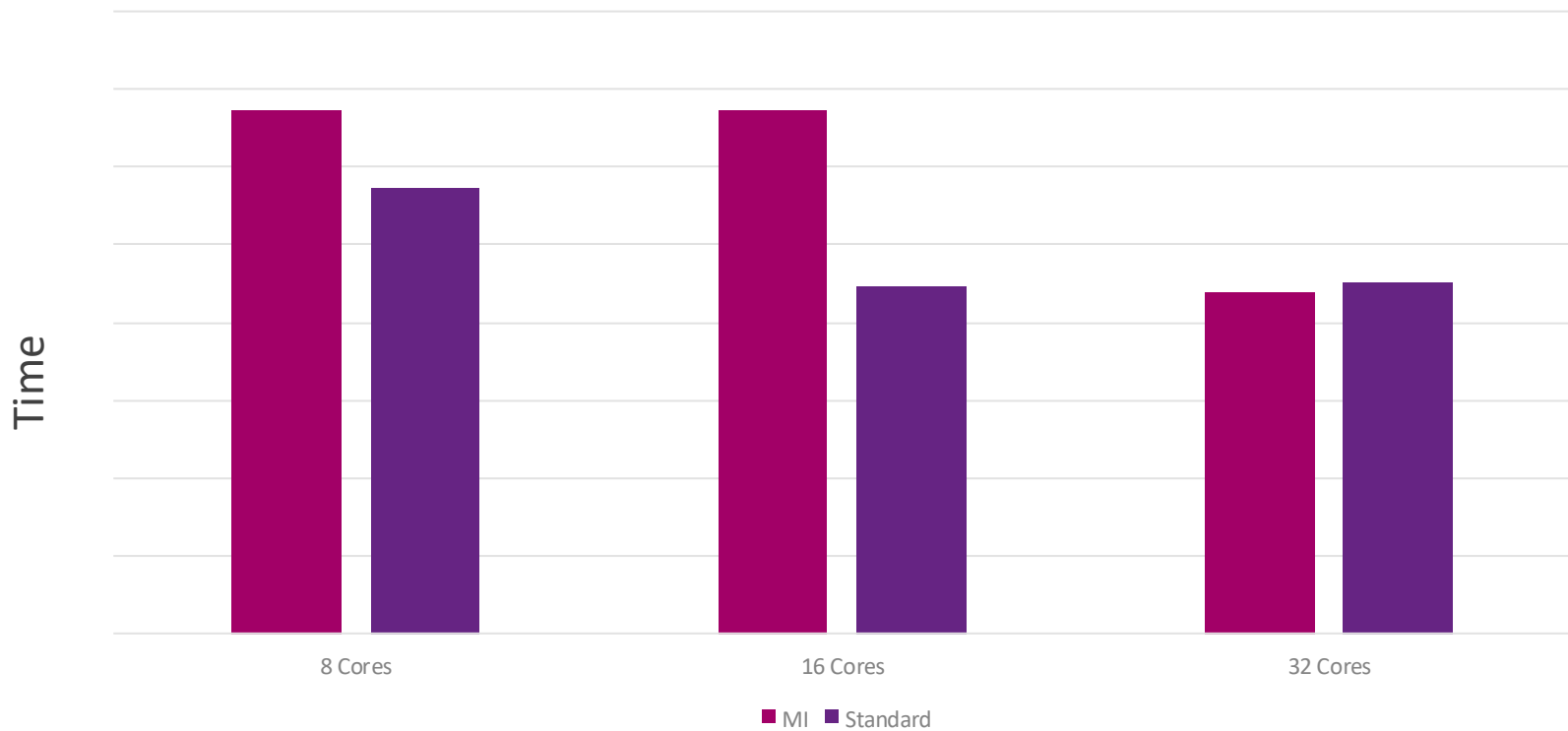
Duration Select 30.000 rows



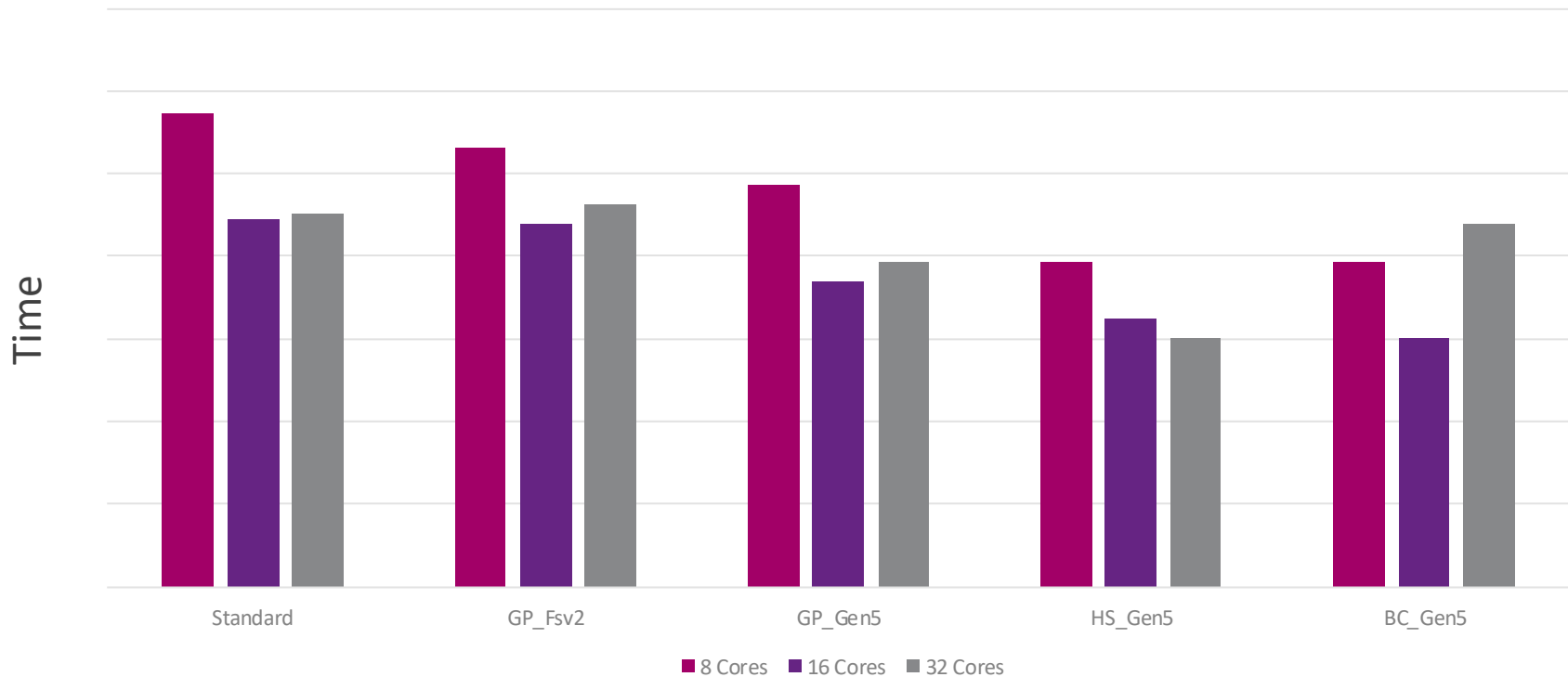
MI Duration 300.000 rows



MI and Standard Duration 300.000 rows



Duration 300.000 rows



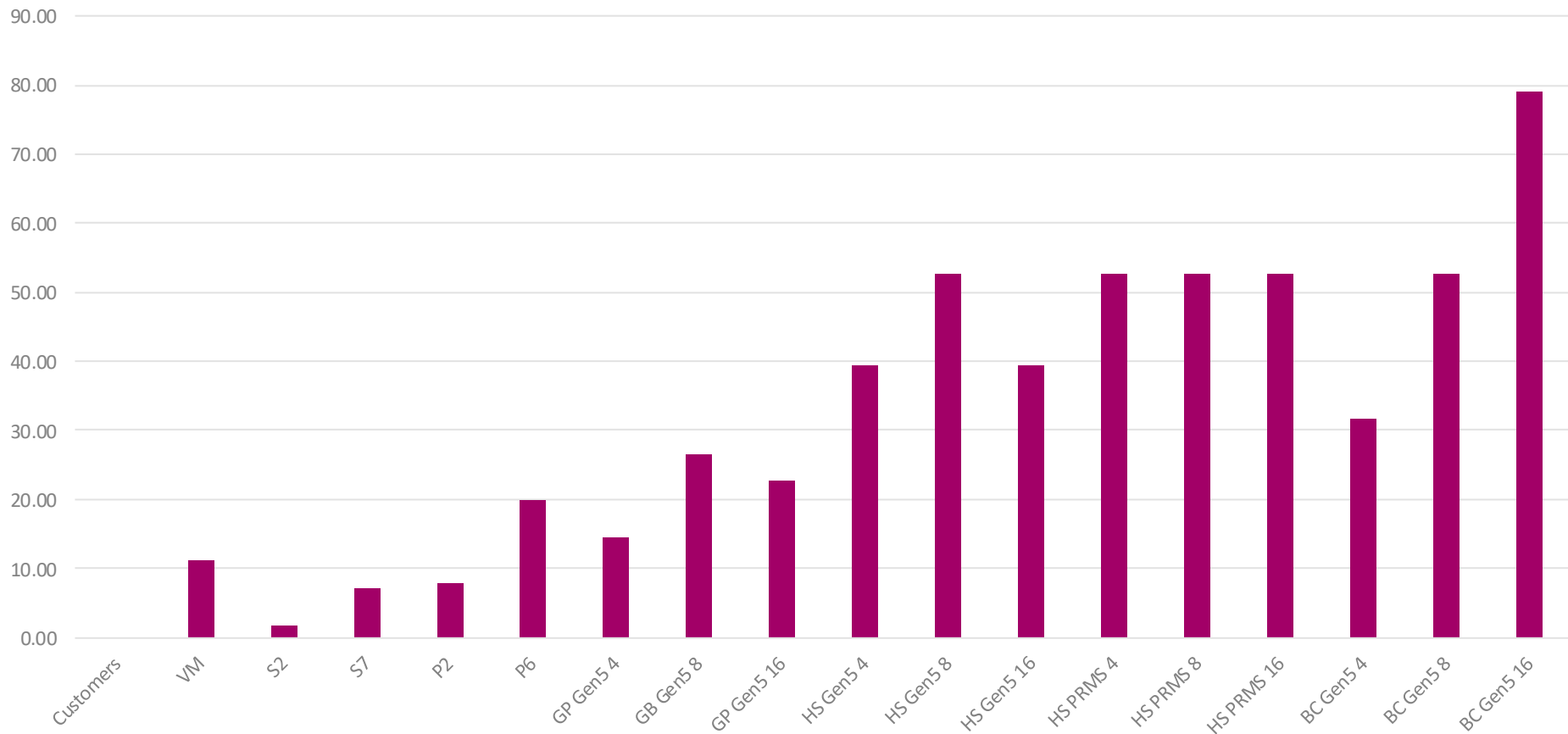
- ▶ Write log
- ▶ Resource Governor
- ▶ IO Queue limit
- ▶ Sosl Scheduler Yield
- ▶ Backup IO
- ▶ Page Latch
- ▶ Latch
- ▶ CXPacket
- ▶ Hadr fabric callback

New Test

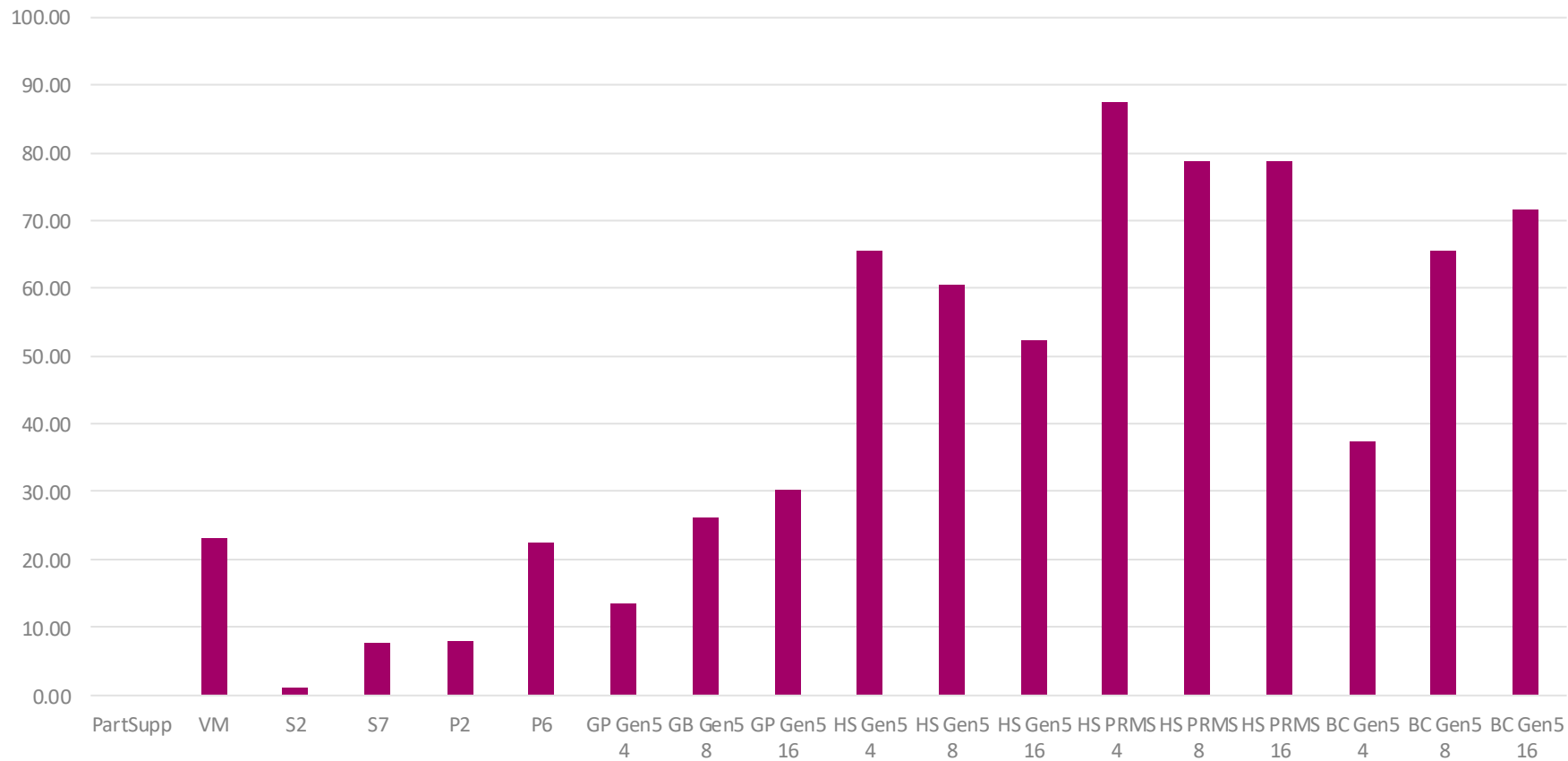
- ▶ Insert CSV Data into Azure SQL
- ▶ TPC-H tables: Customers, PartSupp and LineItem
- ▶ DTU: S2, S7, P2, P6
- ▶ 4, 8 and 16 cores

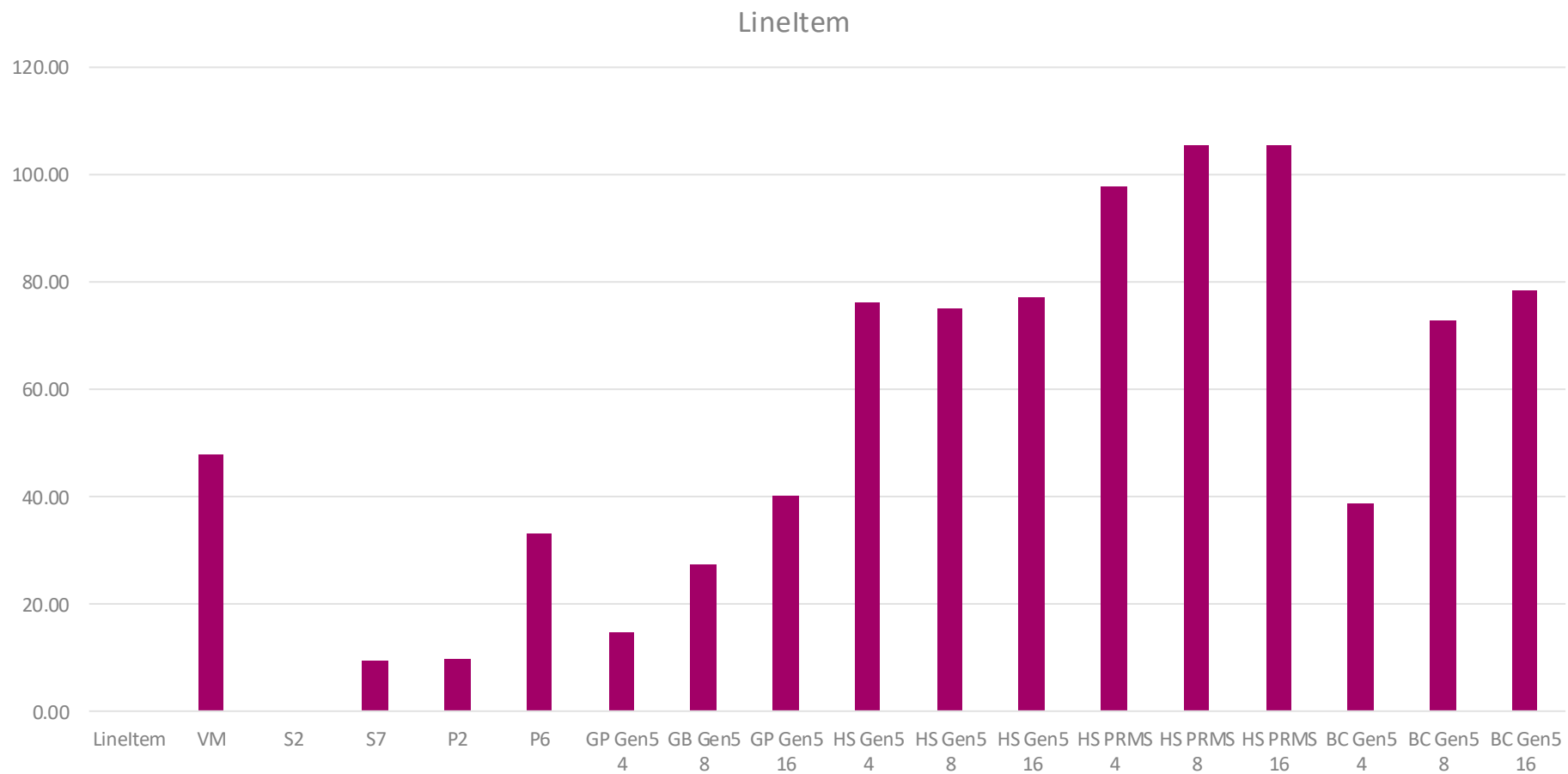
9.2 GB IN 16 FILES

Customers

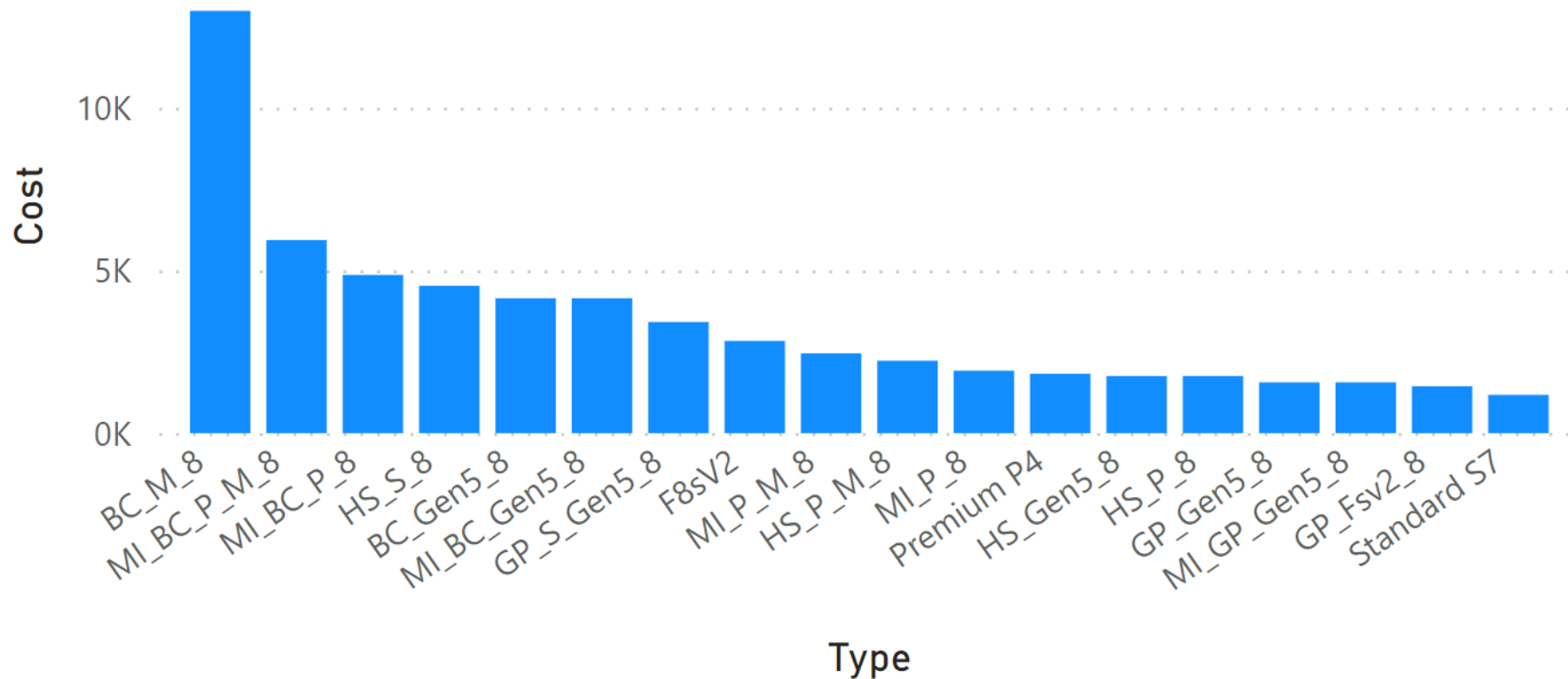


PartSupp

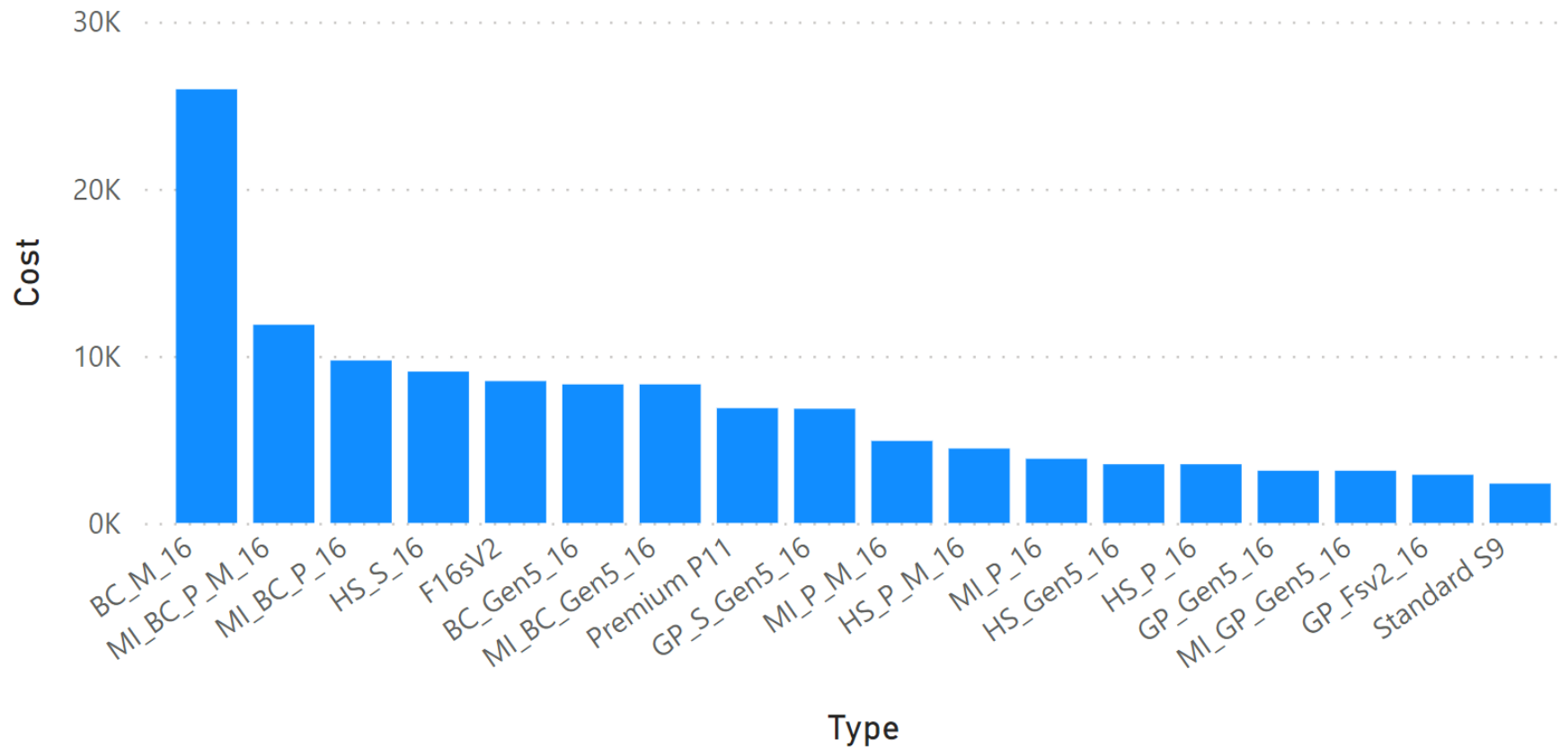




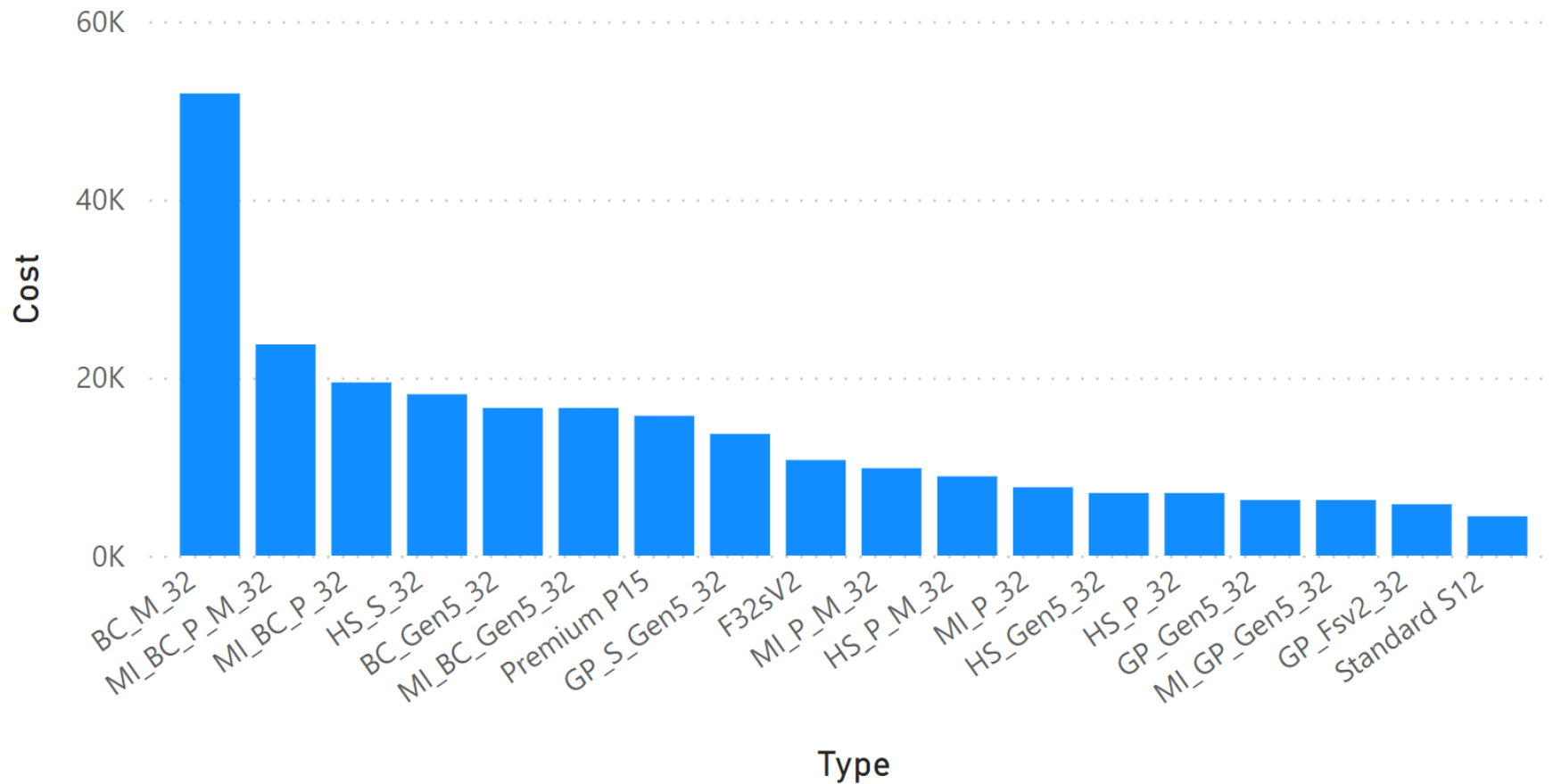
Cost per Type



Cost per Type



Cost per Type



- ▶ If you have intermittent workloads, start with standard or serverless
- ▶ If you have a continuous workload, or at least more than 25% of the time, go provisioned
- ▶ If you need speed, go hyperscale or business critical
- ▶ If you need large databases (over 40 TB), go hyperscale
- ▶ If you have legacy code, go either managed instance or use a VM.
- ▶ Start small and scale when needed





AXA crashtests

AXA.ch

TEST

On a blog site near you

(sshh this is a sneak peak)



INSERT CSV DATA



COMPARING 4-8-16 CORE
OFFERINGS, INCLUDING
AZURE SQL VM



COMPARING INGESTION
TECHNIQUES (BULK INSERT,
POWERSHELL, SSIS)



Azure SQL Databases, tales from the trenches

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<https://github.com/reitse/Speaking>

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Choosing the Azure SQL DB tier, tales from the trenches

Thank you!